MODELING AND ANALYSIS OF IEEE 13 BUS UNBALANCED DISTRIBUTION SYSTEM USING OPENDSS

Mini-Project Report submitted to the SASTRA Deemed to be University in partial fulfillment of the requirements for the award of the degree

B. TECH. ELECTRICAL AND ELECTRONICS ENGINEERING (SMART GRID & ELECTRIC VEHICLES)

Mini-Project Report Submitted by

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Bonafide Certificate

This is to certify that the report titled **"MODELING AND ANALYSIS OF IEEE 13 BUS UNBALANCED DISTRIBUTION SYSTEM USING OPENDSS"** submitted in partial fulfillment of the requirements for the award of the degree of **B. Tech. Electrical & Electronics Engineering** to the SASTRA Deemed to be University, is a bona-fide record of the work done by **ADITHYA A.V. (Reg. No.: 127159002), DINESH GOVIND G.C. (Reg. No.: 127159016), S. AKASH (Reg. No.: 127159004)** during the academic year 2025-26, in the School of Electrical & Electronics Engineering, under my supervision. This report has not formed the basis for the award of any degree, diploma, associateship, fellowship or other similar title to any candidate of any University.

Signature of Project Supervisor:

Name with Affiliation : Dr. D. Karthikaikannan (Sr.Asst.Professor/EEE/SEEE)

Date : 08/05/2026

Project Viva voce held on ____________________

Examiner 1 **Examiner 2**

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Declaration

We declare that the report titled **“MODELING AND ANALYSIS OF IEEE 13 BUS UNBALANCED DISTRIBUTION SYSTEM USING OPENDSS”** submitted by us is an original work done by us under the guidance of **Dr.D. KARTHIKAIKANNAN, Sr.Asst.Professor**, School of Electrical and Electronics Engineering, SASTRA Deemed to be University during the Sixth semester of the academic year 2025-26, in the School of Electrical and Electronics Engineering the work is original and wherever we have used materials from other sources, we have given due credit and cited them in the text of the report. This report has not formed the basis for the award of any degree, diploma, associate-ship, fellowship or other similar title to any candidate of any University.

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Finally, we gratefully acknowledge the encouragement and support of our family and friends, which played a vital role in the successful completion of this project.

ABSTRACT

In real distribution systems, single-phase and two-phase loads from rural and residential consumers create network imbalance, affecting voltage profiles and power losses. This work studies the IEEE 13-bus unbalanced distribution system using OpenDSS to analyze the impact of such loading conditions. Load flow analysis is performed to evaluate per-phase voltage variations, active and reactive power losses, and power flow between buses and transformers. The study compares the IEEE standard benchmark system with modified cases including additional loads and capacitors. Results show significant variations in voltage profiles and system losses under different operating conditions, demonstrating the importance of accurate simulation for understanding distribution system performance.

Specific Contribution

- Modeled IEEE 13-bus system in OpenDSS under base and modified load cases.
- Analyzed voltage profiles, power losses, and power flow in all buses.
- Compared system performance with and without load and capacitors.

Specific Learning

- OpenDSS modeling of unbalanced distribution systems.
- Effect of unbalanced loads on voltage and losses.
- Load flow analysis in practical systems.

Signature of the Guide **Student Reg. No:** 127159004

Name of the Guide: Dr. D. Karthikaikannan (Sr.Asst.Professor/EEE/SEEE)**Name:**AKASH S

ABSTRACT

This work focuses on identifying weak buses in the IEEE 13-bus distribution system for optimal photovoltaic (PV) integration using OpenDSS.Voltage profile analysis is used to determine buses with low per-unit voltage caused by long feeder distances, high line impedance, and heavy loading conditions. Based on this analysis, buses 675, 680, 611, and 652 are selected for PV placement.A solar generation profile is applied to study the impact on system performance. The results show improved voltage levels and reduced power losses due to reduced current flow, with bus 675 showing the most significant improvement compared to other buses.

Specific Contribution

- Studied effect of PV integration on voltage and losses.
- Applied capacitors banks for compensation.
- Compared system performance before and after compensation.

Specific Learning

- Reactive power impact on voltage stability.
- Capacitor role in voltage regulation.
- Loss reduction through compensation.

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Name of the Guide: Dr. D. Karthikaikannan (Sr.Asst.Professor/EEE/SEEE)**Name:** ADHITYA A.V.

ABSTRACT

This study investigates reactive power compensation in the IEEE 13-bus distribution system by introducing additional loads and analyzing their impact on voltage profiles and power losses. Increased loading leads to higher reactive power demand, resulting in voltage drops and increased system losses. To address this, shunt capacitors are considered for compensation to improve voltage regulation and reduce reactive power flow from the source. Load flow analysis is used to evaluate system performance before and after compensation. The results demonstrate improved voltage stability and reduced losses, highlighting the effectiveness of capacitor placement in enhancing distribution system performance.

Specific Contribution

- Identified weak buses (675, 680, 611, 652) using voltage profile analysis.
- Studied causes like line impedance, feeder length, and heavy loading.
- Evaluated PV impact on voltage and losses.

Specific Learning

- Voltage profile analysis for weak bus detection.
- Role of PV in voltage improvement.
- Optimal placement of distributed generation.

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NOTATIONS

P	Active power (kW)
Q	Reactive power (kVAR)
S	Apparent power (kVA)
P_{peak}	Peak load power (kW)
P_{min}	Minimum load power (kW)
$P(t)$	Load at time t (kW)
P_{PV}	Photovoltaic power (kW)
$P_{PV,max}$	Maximum allowable PV capacity (kW)
$P_{675a}, P_{675b}, P_{675c}$	Phase-wise active power at Bus 675
Q_c	Capacitor reactive power (kVAR)
PF	Power factor
ϕ	Power factor angle
$\tan\phi_1$	Initial power factor angle
$\tan\phi_2$	Target power factor angle
V	Bus voltage (p.u)
I	Line current (A)
Z	Line impedance
V_{spec}	Specified voltage
V_{new}	Updated voltage
$mult(t)$	Load multiplier at time t
ϵ	Convergence tolerance
DG	Distributed Generation
PV	Photovoltaic system
PQ	Constant power mode
PV mode	Constant voltage mode

ABBREVIATIONS

OpenDSS	Open Distribution System Simulator
PV	Photovoltaic System
DG	Distributed Generation
MPP	Maximum Power Point
MPPT	Maximum Power Point Tracking
ANSI	American National Standards Institute
I^2R	Resistive Power Loss
VUF	Voltage Unbalance Factor

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CHAPTER 1

INTRODUCTION

1.1 PROJECT OVERVIEW

Load flow analysis of the IEEE 13-bus system is performed to evaluate voltage profile (p.u) and system losses under normal conditions. Load variation is introduced to study its impact on voltage stability and losses. Capacitor banks are used for reactive power compensation to improve voltage profile and reduce I^2R losses. Photovoltaic (PV) generation is integrated at weak buses to analyze voltage profile improvement and loss reduction. Comparative analysis of selected buses is carried out to determine optimal PV placement. Results show improved efficiency and reliability through load variation, compensation, and distributed generation.

1.2 IEEE 13-BUS SYSTEM OUTLINE

The IEEE 13-bus is an unbalanced radial distribution feeder modeled in OpenDSS. It operates at 4.16 kV and is supplied from a 115 kV source through a 5 MVA step-down transformer between the substation and primary feeder.

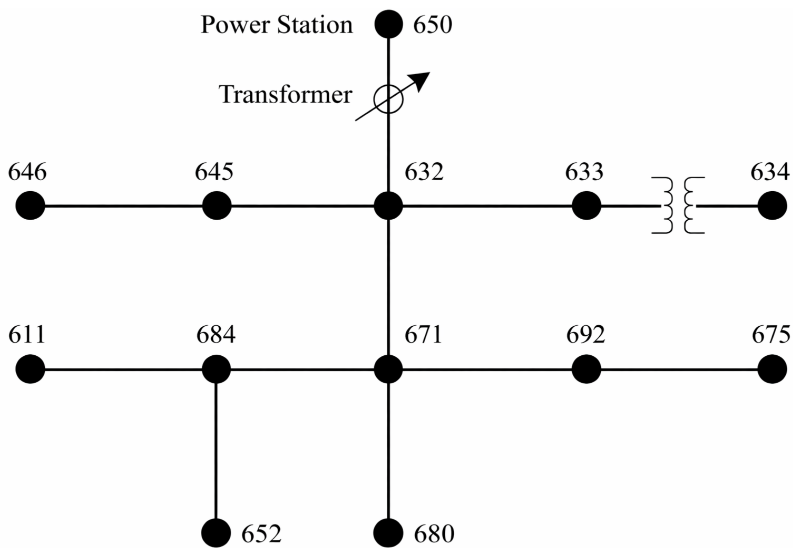

Figure 1.1: IEEE 13-Bus distribution feeder

The system includes three-phase, two-phase, and single-phase lines, loads, voltage regulators, and a distribution transformer, representing a practical rural unbalanced network.

1.3 REPORT LAYOUT

This provides a systematic structure to present the modeling, analysis which includes introduction,literature review,methodology,load alteration and PV integration.

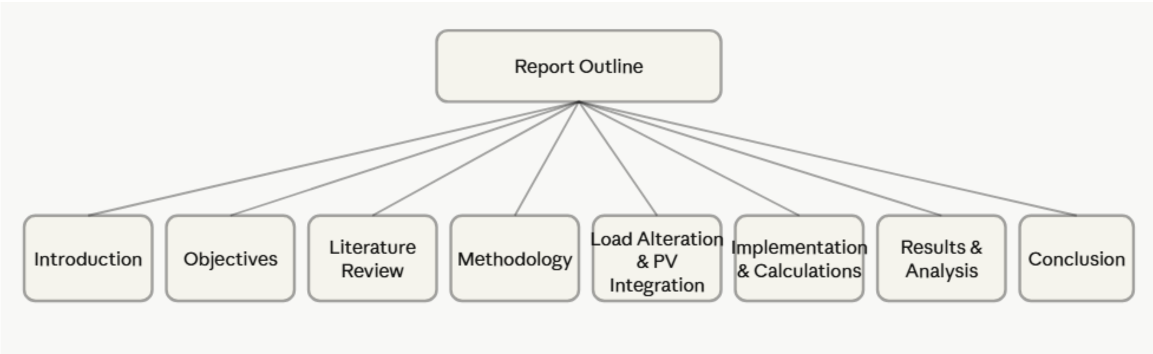

Figure 1.2: REPORT LAYOUT

1.3.1 Introduction

Introduction to the project with an overview of the IEEE 13-bus system and objectives, including analysis under varying load conditions, impact of PV on voltage profile and losses, effectiveness of capacitor compensation, identification of optimal PV placement, and coordinated strategies for voltage regulation and loss reduction.

1.3.2 Literature Review

Discusses prior research on unbalanced distribution systems, distributed generation, and PV integration, highlighting research gap and motivation.

1.3.3 Methodology

Describes system modeling, load variation, capacitor compensation, PV integration, and workflow.

1.3.4 Load Alteration and PV Integration

Focuses on load variation and PV placement with impact analysis.

1.3.5 Implementation and Calculations

Includes OpenDSS modeling, load distribution, capacitor sizing, PV sizing, and time-varying simulations.

1.3.6 Results and Comparison

Compares voltage profile and losses for base case, load addition, capacitor, and PV cases.

1.3.7 Conclusion

Summarizes improvements in voltage profile and loss reduction, emphasizing coordinated PV and reactive power compensation.

CHAPTER 2

LITERATURE REVIEW

2.1 CASE STUDY

This section reviews three papers on PV integration in distribution systems, highlighting optimal placement, loss reduction, and voltage stability in unbalanced networks.

2.1.1 Impact Assessment of PV Penetration on Unbalanced Distribution network with Dynamic load condition

Algorithm and Methodology

An integrated approach using OpenDSS and MATLAB is employed, where MATLAB is interfaced with OpenDSS to perform 24-hour dynamic simulations. The IEEE 13-bus system data is used to establish a time-based simulation in OpenDSS for load flow analysis. MATLAB is then used for time-series evaluation of system losses, voltage profile, and voltage unbalance. Photovoltaic systems are injected at different buses, and the results are analyzed to determine optimal placement and sizing.

PV Placement Strategy and Results

Photovoltaic capacities of 100 kW, 200 kW, and 250 kW are tested at buses 611, 675 (with a capacitor), and 652 (without a capacitor). The results show that minimum line losses occur at bus 675, while bus 652 exhibits approximately 20 percent higher losses due to reverse power flow. The optimal PV size is found to be 100 kW, as higher penetration levels increase losses, with peak losses occurring around 13:00 hours. Voltage unbalance, defined as the ratio of negative to positive sequence components, remains within the range of 0.5 percent to 2.5 percent at bus 675 with 100 kW PV. In the absence of PV, the maximum unbalance occurs at bus 675 during peak load conditions. PV integration at bus 675 improves voltage stability and prevents overvoltage and voltage collapse during peak demand. Additionally, bus 633 exhibits the maximum voltage, while bus 632 shows the minimum voltage.

2.1.2 Analysis of IEEE-13 Bus System with PV and Battery

The IEEE 13-bus system is analyzed under different scenarios, including without PV and battery, and with the integration of solar PV and battery storage. The system is modified by adding a generator and a transformer at bus 675, along with load connections. The analysis is carried out using MATLAB and OpenDSS. Loads are connected at a voltage level of 4.16 kV, and Phase A carries approximately 57 percent of the total load, indicating a highly unbalanced system.

Cases Analyzed and Results

Case	Phase A	Phase B	Phase C
Without DG	Normal	Slightly Higher	Normal
With DG (PV)	Increased-0.698 %	Increased-0.117 %	Increased-0.558 %
With DG and Load Variation	Decreased-0.336 %	Decreased-0.058 %	Decreased-0.773 %
With Battery	Decreased-0.557 %	Increased-0.543 %	Increased-0.180 %

Table 2.1: Voltage Variation for Different Operating Conditions

Observation

- Photovoltaic integration increases voltage in one phase by 0.698 percent while decreasing it by 0.773 percent in another phase, indicating system unbalance.
- Battery operation causes a slight voltage decrease of approximately 0.577 percent in the phase where PV had increased the voltage, contributing to voltage regulation.

2.1.3 Unbalanced Distribution Power Flow with Distributed Generation

Node Renumbering

Node numbering is performed using a breadth-first approach, where lateral branches are indexed before returning to the main feeder.

Backward–Forward Sweep

The forward sweep calculates bus voltages based on line impedance and current flow from downstream buses, while the backward sweep updates bus voltages using source voltage conditions and calculated currents. The iterative process continues until the difference between calculated and specified voltages satisfies the convergence criterion.

Case Study Results

- **Base Case:** The system converges in three iterations and matches the results obtained using RDAP analysis. The minimum voltage observed is 0.8967 per unit at node 611.
- **Distributed Generation as PQ Node:** With power injection of 630 kilowatts per phase and reactive power of 305 kilovolt-ampere reactive per phase, the source current reduces significantly from approximately 592 amperes to 274 amperes, and the voltage profile improves.
- **Distributed Generation as PV Node:** The voltage at node 671 is maintained at 1.0 per unit with reactive power support of 1.437 megawatts for the three-phase system. The voltage at node 675 improves from 0.9168 per unit to 0.9807 per unit.

The results demonstrate the significant impact of distributed generation on voltage profile, current flow, and system performance in an unbalanced distribution network.

2.2 RESEARCH GAP

1. Most studies treat PV, battery storage, and capacitor compensation independently, without analyzing their coordinated impact under varying load conditions and DG operating modes (PQ/PV) in unbalanced networks.
2. The combined effect of PV penetration and reactive power compensation under dynamic loading is insufficiently explored, particularly for achieving simultaneous voltage stability, loss reduction, and imbalance mitigation.
3. There is a lack of adaptive PV sizing and optimal bus selection strategies that consider real-time load variation, system impedance, and network topology, limiting effective integration and overall system performance.

CHAPTER 3

OBJECTIVES

1. To carry out load flow analysis on the IEEE 13-bus system in order to understand the voltage profile,power losses under normal operating conditions.
2. To understand the impact of increasing load demand on the system by adding extra loads and observing how it affects power losses and voltage stability.
3. To evaluate the effectiveness of capacitor placement in reducing power losses and improving the voltage profile of the system.
4. To integrate photovoltaic (PV) generation into different buses and examine its influence on system performance, including loss reduction and voltage improvement.
5. To identify the most suitable location for PV installation by comparing its impact on various buses and selecting the bus for PV placement which shows maximum efficiency.

CHAPTER 4

ANALYSIS-WORKFLOW

1. OpenDSS - system modelling load flow analysis implementation with standard IEEE data
2. Load Addition -Single phase ,two phase and three phase loads like Residential, agricultural, and commercial loads added to increase the unbalance in loads.
3. Capacitor Compensation - Reactive power support for added loads.
4. 24-hour load profile employed.
5. PV Integration - Solar PV with 10% penetration limit for safe penetration and over-voltage protection.
6. Capacitor added - Reactive Power Compensation for PV for Q-power support
7. 24 hrs-load and solar profiles employed.
8. Monitoring and Result Analysed

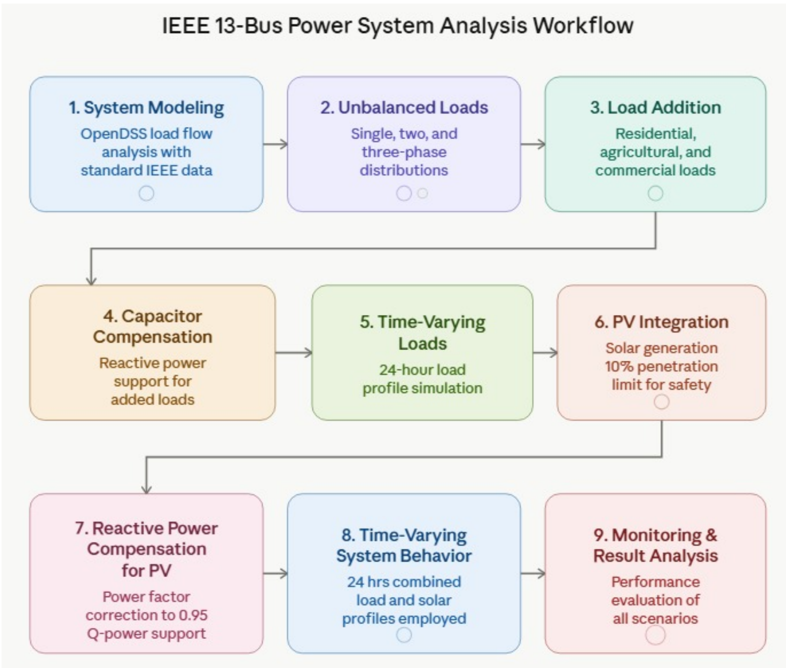

Figure 4.1: ANALYSIS WORK FLOW

CHAPTER 5

LOAD VARIATION AND SOLAR PV INTEGRATION

5.1 LOADS ALTERATION

LOADS	Bus Connected	Phases	Connection	P (kW)	Q (kVAR)
Irrigation Pump - motor load	671.1	1	Wye	80	40
Residential loads- Fans, lights, AC	680.1.2.3	3	Wye	150	90
Cooling systems (AC) - Schools	684.3	1	Wye	60	30
Agro Processing Mill	692.1.2.3	3	Wye	120	70

5.2 CAPACITOR ADDITION

CAPACITORS	Bus Connected	Phases	Reactive Power (kVAR)
Cap3	671.1	1	40
Cap4	680	3	100
Cap5	684.3	1	30
Cap6	692	3	80

Observation

- Due to increment in loads it results in more current absorption will take place.
- To balance the current losses compensation capacitors were added at subsequent
- places

5.3 PV INTEGRATION

Reason for Selection of Buses for PV integration

BUS 675

- Connected through bus 692 via a three-phase line, forming a downstream section of the feeder.
- Carries significant single-phase loads across all phases (Load.675a, 675b, 675c), causing phase imbalance.
- Electrically distant from main feeder path (650–632–671), resulting in higher impedance and voltage drop.
- PV integration provides local power injection, reduces upstream current, and improves voltage profile.

BUS 680

- Fed from bus 671 through a short three-phase line (Line.671680), close to feeder backbone.
- No major local load, but influenced by upstream loading at bus 671.
- Voltage variations governed by upstream load flow and regulator action at bus 650.
- PV placement helps analyze reverse power flow and voltage rise near feeder head.

BUS 611

- Supplied from bus 684 through a single-phase lateral (Line.684611).
- Connected to a single-phase load (Load.611) and supported by a capacitor (Cap2).
- Experiences low current and minimal voltage drop compared to deeper buses.
- PV integration gives limited improvement due to low demand and existing compensation.

BUS 652

- Located at the end of a long single-phase lateral (Line.684652), making it electrically weak.
- Connected via high-impedance line (mtx607), causing significant voltage drop.
- Supplies a single-phase load (Load.652) with limited reactive support.
- PV improves voltage regulation and reduces upstream feeder losses.

Observation of PV Placement Analysis

Weak buses are identified based on distribution system topology, impedance, and load distribution.

The end buses (675 and 652) show higher voltage drops and benefit most from PV integration compared to other weak buses.

Upstream buses (680 and 611) show limited improvement due to lower impedance paths or existing compensation.

Proper placement of PV enhances the voltage profile, reduces losses, and improves overall system performance.

CHAPTER 6

MODELING AND CALCULATION

6.0.1 System Modeling in OpenDSS

The system is modeled in OpenDSS using standard IEEE data.Line between the buses are specified using resistance and reactance matrices (unequal phase impedances) are considered to accurately represent unbalanced operating conditions.

Unbalance Nature of Loads

The IEEE 13-bus system contains single,two and three phase loads which creates phase imbalance, voltage deviation, and increased feeder losses, which are crucial in rural systems.

Table 6.1: Load Distribution Across Buses

Bus	**Phase**	**Connection**	**kW**	**kVAR**
671	1,2,3	Delta	1155	660
634	1	Wye	160	110
634	2	Wye	120	90
634	3	Wye	120	90
645	2	Wye	170	125
646	2-3	Delta	230	132
692	3-1	Delta	170	151
675	1	Wye	485	190
675	2	Wye	68	60
675	3	Wye	290	212
611	3	Wye	170	80
652	1	Wye	128	86

6.0.2 Analysis with addition of loads and capacitor compensation

Additional realistic loads representing residential, agricultural and commercial demand are incorporated into the system to reflect practical rural distribution conditions.

Table 6.2: Additional Load Modeling in OpenDSS

Load Name	Bus	Phase	Conn.	kW	kVAR
IrrigationPump	671.1	1	Wye	80	40
Residential	680.1.2.3	3	Wye	150	90
CoolingSchools	684.3	1	Wye	60	30
AgroProcessing	692.1.2.3	3	Wye	120	70

6.0.3 Daily Load Shape Profile added

The 24-hour load variation used in the simulation is defined as follows:

Table 6.3: 24-Hour Load Shape Multipliers

Hour	Multiplier	Hour	Multiplier	Hour	Multiplier
1	0.60	9	1.20	17	1.30
2	0.55	10	1.30	18	1.25
3	0.50	11	1.40	19	1.20
4	0.50	12	1.45	20	1.10
5	0.55	13	1.50	21	1.00
6	0.70	14	1.45	22	0.90
7	0.90	15	1.40	23	0.80
8	1.10	16	1.35	24	0.70

6.0.4 Capacitor Sizing Calculations for Compensation Buses

Capacitor banks are designed to improve power factor from existing condition to a target value of 0.95.

$$\mathrm{PF}_{target} = 0.95 \quad \Rightarrow \quad \tan\phi_2 = 0.328$$

$$Q_c = P(\tan\phi_1 - \tan\phi_2), \quad \tan\phi_1 = \frac{Q}{P}$$

Bus 671.1 (Irrigation, 2.4 kV)

$$Q_c = P_{671.1}\left(\frac{Q_{671.1}}{P_{671.1}} - 0.328\right) = 40\ kVAR$$

Bus 680 (Residential, 4.16 kV)

$$Q_c = P_{680}\left(\frac{Q_{680}}{P_{680}} - 0.328\right) = 100\ kVAR$$

Bus 684.3 (School Load, 2.4 kV)

$$Q_c = P_{684.3}\left(\frac{Q_{684.3}}{P_{684.3}} - 0.328\right) = 30\ kVAR$$

Bus 692 (Agro-processing, 4.16 kV)

$$Q_c = P_{692}\left(\frac{Q_{692}}{P_{692}} - 0.328\right) = 80\ kVAR$$

Table 6.4: Capacitor Compensation Summary for IEEE 13-Bus System

Bus	Load Type	Voltage (kV)	Capacitor (kVAR)
671.1	Single-phase Irrigation	2.4	40
680.1.2.3	Three-phase Residential	4.16	100
684.3	Single-phase School	2.4	30
692	Three-phase Agro-processing	4.16	80

6.0.5 Load flow analysis with PV integration

A base case load flow is performed without PV or additional compensation to obtain reference values of bus voltages (per unit), line currents, and system losses for comparison.

6.0.6 PV System Sizing and Integration

Load Data at Bus 675

The total load at Bus 675 is obtained from three single-phase loads:

$$P_{675a} = 485 \text{ kW}$$

$$P_{675b} = 68 \text{ kW}$$

$$P_{675c} = 290 \text{ kW}$$

Total Peak Load:

$$P_{\text{peak}} = 485 + 68 + 290$$
$$P_{\text{peak}} = 843 \text{ kW}$$

24-Hour Load Variation Model

The time-varying load is modeled using the given multiplier profile:

$$P(t) = P_{\text{peak}} \times \text{mult}(t)$$

Where:

- $P(t)$ = load at time t
- $\text{mult}(t)$ = hourly load multiplier

Minimum Load Calculation

From the given 24-hour profile, the minimum multiplier is:

$$\text{Minimum multiplier} = 0.40$$

Therefore:

$$P_{\text{min}} = P_{\text{peak}} \times 0.40$$
$$P_{\text{min}} = 843 \times 0.40$$
$$P_{\text{min}} = 337.2 \text{ kW}$$

PV Sizing Constraint (Minimum Load Method)

The PV penetration is limited by:

$$P_{\text{PV}} \leq P_{\text{min}}$$

This ensures prevention of reverse power flow and overvoltage conditions at the bus.

Maximum PV Capacity at Bus 675

$$P_{\text{PV,max}} \approx 337.2 \text{ kW}$$

Thus, the PV capacity is selected as 300 kW, which is close to the calculated limit.

6.0.7 Capacitor Sizing and Implementation

The capacitor ratings are calculated based on power factor correction to 0.95.

Target Power Factor: $PF = 0.95, \quad \tan\phi = 0.328$

- **Bus 675:** $P \approx 485\ kW$, initial $Q \approx 190\ kVAR$

$$Q_c = P\tan\phi - Q \Rightarrow \text{Capacitor} \approx 310\ kVAR$$

- **Bus 611:** $P \approx 170\ kW$, initial $Q \approx 80\ kVAR$

$$Q_c \approx 24\ kVAR$$

Table 6.5: Additional Capacitor Placement in the IEEE 13-Bus System

Capacitor	**Bus**	**Phase**	**kVAR**	**kV**
Cap1	675	3	300	4.16
Cap2	611.3	1	30	2.4

6.0.8 Time-Varying System Behavior

A 24-hour time-series simulation is conducted using load and solar profiles.

Table 6.6: 24-Hour Load and Solar Shape Multipliers

Hr	1	2	3	4	5	6	7	8	9	10	11	12
Load	0.60	0.55	0.50	0.45	0.40	0.45	0.60	0.75	0.90	1.00	1.10	1.15
PV	0.00	0.00	0.00	0.00	0.00	0.05	0.20	0.40	0.60	0.80	0.90	0.95
Hr	13	14	15	16	17	18	19	20	21	22	23	24
Load	1.20	1.25	1.20	1.15	1.10	1.20	1.35	1.50	1.40	1.20	0.95	0.75
PV	0.90	0.85	0.75	0.55	0.35	0.15	0.05	0.00	0.00	0.00	0.00	0.00

CHAPTER 7

RESULT COMPARISON AND FUTURE SCOPE

7.1 MONITORING, PERFORMANCE EVALUATION AND RESULT ANALYSIS

Bus voltages and power losses from various conditions like base case, load addition with capacitor compensation, PV integration with capacitor compensation.Results of voltage profiel and power losses produced are discussed below:

7.2 LOAD ALTERATION AND COMPENSATION

7.2.1 Voltage Comparison

● Phase A ● Phase B ● Phase C

Before Addition of New Loads with Capacitor Compensation

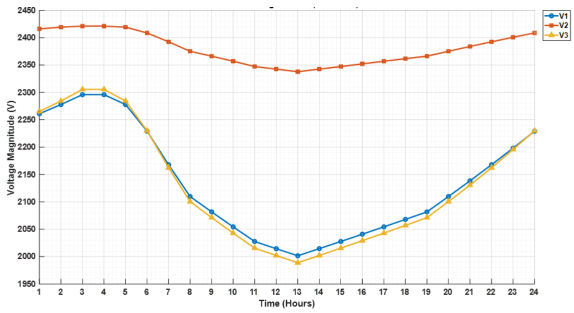

Figure 7.1: Voltage profile before addition of new loads with capacitor compensation

After Addition of New Loads with Capacitor Compensation

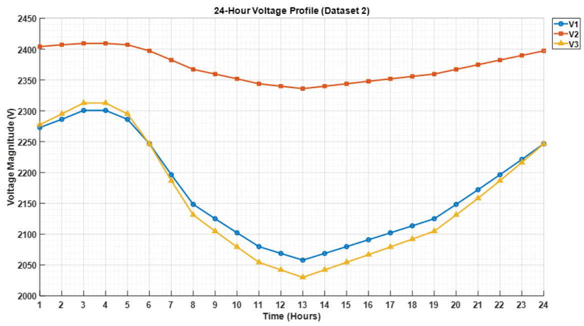

Figure 7.2: Voltage profile after addition of new loads with capacitor compensation

7.2.2 Losses Comparison

Loss for base case

CATEGORY	POWER LOSS (kW)
Line Losses	48.5
Transformer Losses	33.4
Total Losses	**81.9**
Total Load Power	**2245.9**
Loss Percentage	3.64 %

Loss after loads added and capacitor compensation

CATEGORY	POWER LOSS (kW)
Line Losses	59.7
Transformer Losses	40.1
Total Losses	**99.8**
Total Load Power	**2512.5**
Loss Percentage	3.97 %

7.3 PV INTEGRATION

7.3.1 Phase Voltage Impact on Bus 675

• BEFORE PV INJECTION • AFTER PV INJECTION

Phase A

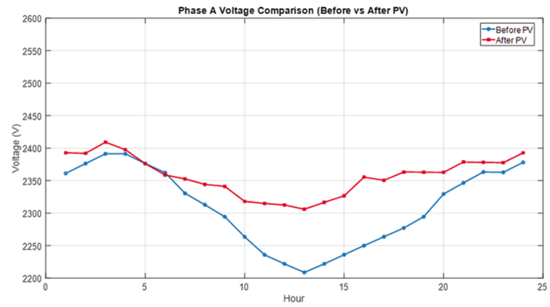

Figure 7.3: Phase A Voltage Profile at Bus 675 (PV Integration)

Phase B

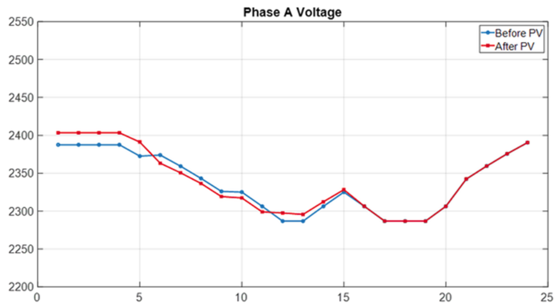

Figure 7.4: Phase B Voltage Profile at Bus 675 (PV Integration)

Phase C

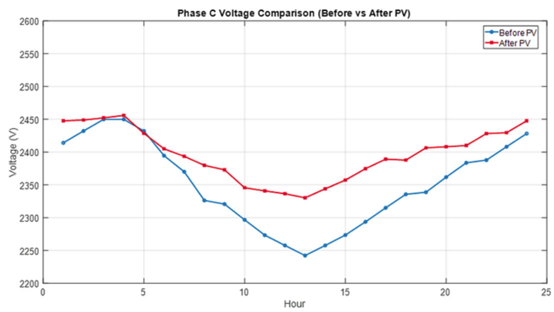

Figure 7.5: Phase C Voltage Profile at Bus 675 (PV Integration)

7.3.2 Phase Voltage Impact on Bus 680

- BEFORE PV INJECTION
- AFTER PV INJECTION

Phase A

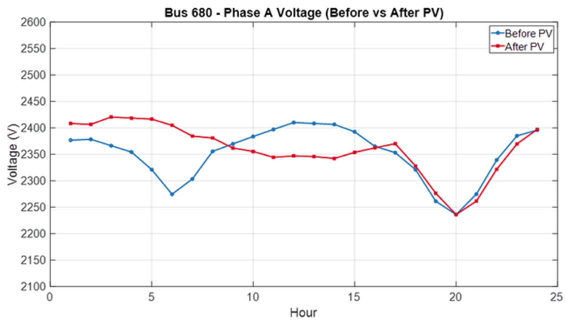

Figure 7.6: Phase A Voltage Profile at Bus 680 (PV Integration)

Phase B

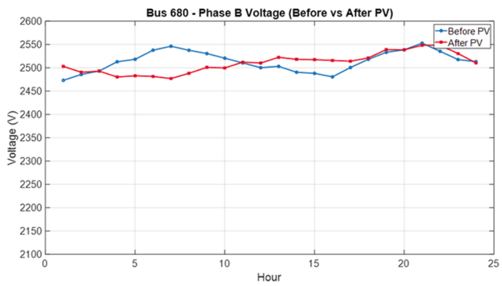

Figure 7.7: Phase B Voltage Profile at Bus 680 (PV Integration)

Phase C

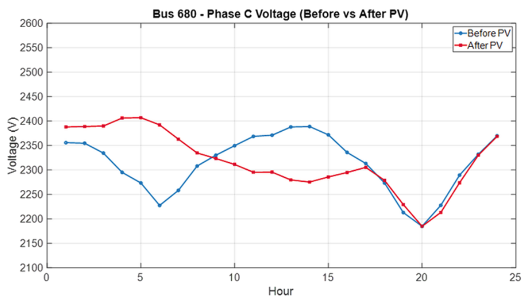

Figure 7.8: Phase C Voltage Profile at Bus 680 (PV Integration)

7.3.3 Phase Voltage Impact on Bus 611

• BEFORE PV INJECTION • AFTER PV INJECTION

Phase C

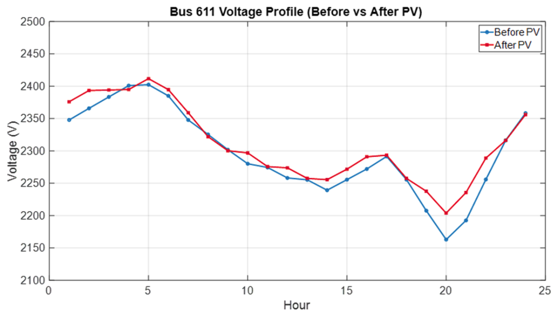

Figure 7.9: Phase C Voltage Profile at Bus 611 (PV Integration)

7.3.4 Phase Voltage Impact on Bus 652

Phase A

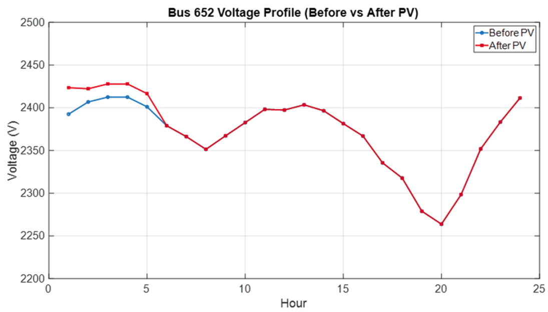

Figure 7.10: Phase A Voltage Profile at Bus 652 (PV Integration)

CHAPTER 8

CONCLUSION

The study concluded that load addition in the IEEE 13-bus unbalanced distribution system increased feeder current demand, causing voltage drops and higher active and reactive power losses. Unbalanced loading significantly affected voltage profile and current distribution across the network. Capacitor compensation improved voltage stability and reduced feeder losses by supplying reactive power locally and reducing reactive power flow from the source.

The study further concluded that PV integration at weak buses such as Bus 675 and Bus 652 improved voltage profile and reduced system losses through local power support and reduced upstream current flow. Electrically distant buses showed greater improvement due to higher sensitivity to voltage variations and line impedance effects. Controlled PV penetration along with capacitor compensation also helped prevent reverse power flow and overvoltage conditions.

Overall, the work concluded that coordinated integration of capacitor compensation and distributed PV systems is effective for improving voltage regulation, reducing feeder losses, and enhancing the performance of unbalanced rural distribution systems. The analysis also demonstrated the effectiveness of OpenDSS as a reliable tool for modeling, analysis, and optimization of modern distribution networks under varying operating conditions.

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8.1 FUTURE SCOPE

PV SIZING METHOD USING MINIMUM LOAD MULTIPLIER

Determination of Peak Load

The total peak load at the selected bus is obtained by summing all connected loads.

$$P_{\text{peak}} = \sum P_{\text{loads}}$$

24-Hour Load Variation Modeling

The daily load variation is represented using hourly load multipliers. The load at any time instant is calculated as:

$$P(t) = P_{\text{peak}} \times \text{mult}(t)$$

where $P(t)$ is the load at time t, P_{peak} is the peak load, and $\text{mult}(t)$ is the hourly load multiplier.

Minimum Load Calculation

The minimum load condition is obtained using the minimum multiplier from the 24-hour load profile.

$$P_{\text{min}} = P_{\text{peak}} \times \text{mult}_{\text{min}}$$

subsubsectionPV Penetration Constraint

The PV penetration level is limited such that the PV generation does not exceed the minimum load demand.

$$P_{\text{PV}} \leq P_{\text{min}}$$

This constraint helps to avoid reverse power flow and overvoltage conditions in the distribution system.

Maximum PV Capacity Selection

The maximum allowable PV capacity is selected approximately equal to the minimum load value.

$$P_{\text{PV,max}} \approx P_{\text{min}}$$

CHAPTER 9

REFERENCES

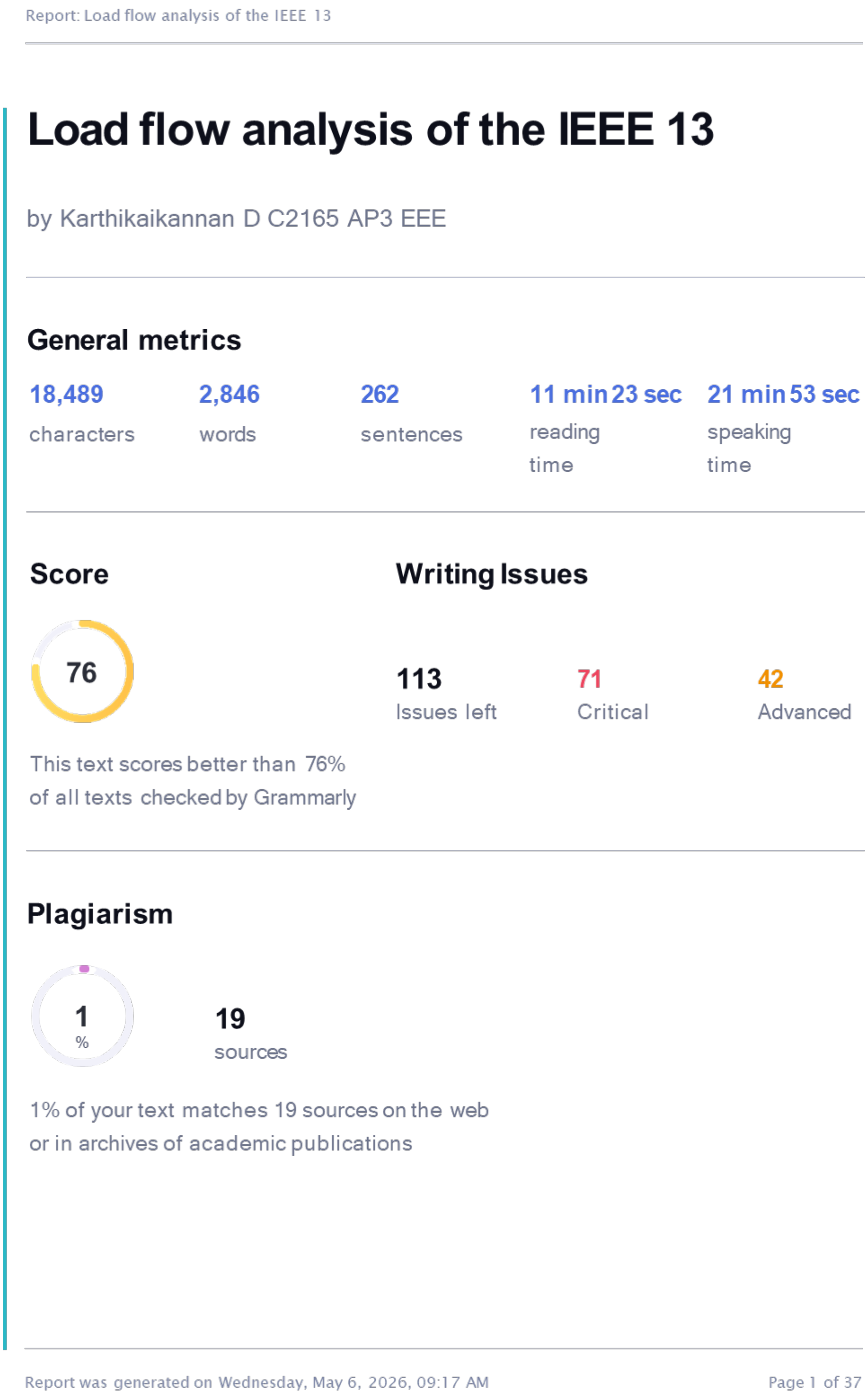
[1] IEEE13 – Bus Analysis in OpenDSS with PV System and Storage.

[2] S. Khushalani and N. Schulz, “Unbalanced Distribution Power Flow with Distributed Generation,” *IEEE/PES Transmission and Distribution Conference and Exhibition*, pp. 301–306, 2006.

[3] “Impact Assessment of PV Penetration on Unbalanced Distribution Network with Dynamic Load Condition,” IEEE.

CHAPTER 10

SIMILARITY CHECK REPORT



Report: Load flow analysis of the IEEE 13

Load flow analysis of the IEEE 13

by Karthikaikannan D C2165 AP3 EEE

General metrics

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Plagiarism

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19 sources

1% of your text matches 19 sources on the web or in archives of academic publications

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Figure 10.1: Plagarism Report